

Therefore the stress ratios are,

Pressure	M / Maxo2	= 2.07 / 5.51	= 0.37	OK
	V / Vay	= 1.38 / 22.01	= 0.06	OK
Suction	M / Maxo	= 2.25 / 3.58	= 0.63	OK
	V / Vay	= 1.50 / 22.01	= 0.06	OK

Check Deflection,

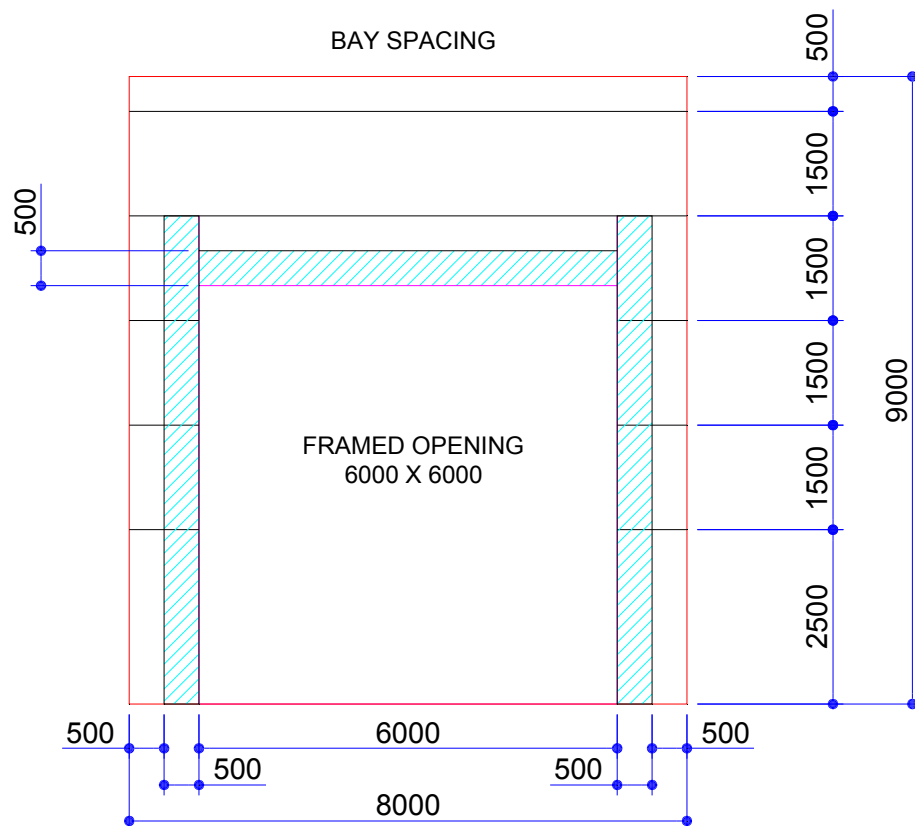
$$I_{xx} = 3.08 \times 10^6 \text{ mm}^4$$

Calculated deflection, suction

$$= 5 \times 0.50 \times 1000 \times 6000^4 / (384 \times 205000 \times 3.08 \times 10^6 \times 1000)$$

$$= 13.36 \text{ mm}$$

$$\text{Allowable deflection} = \text{span} / 90 = 6000 / 90 = 66.67 \text{ mm} \quad \text{OK}$$



Design of Jambs

For the design of jambs it is important to note the connection detail of the opening purpose, such as roll up door, sliding door or open for access. The tributary width in all cases will be different,

Tributary width for Roll Up Door

$$= (\text{Framed Opening Width} / 2 + 0.5) \\ = (6 / 2 + 0.5) = 3.5 \text{ m}$$

Tributary width for Sliding Door or open for access = 0.5 m

a) Considering, the opening for a Roll-Up Door

Effective wind load area is the greater of,

$$\text{Tributary area} = 3.5 \times 7 = 24.5 \text{ m}^2$$

$$\text{Span} \times \text{Span} / 3 = 7 \times 7 / 3 = 16.3 \text{ m}^2$$

$$\text{Hence effective wind load area} = 24.5 \text{ m}^2$$

Considering an enclosed building and roof slope less than 10° , referring table 1.4.6 (a) of MBMA 02, GCp for interior zone is given by,

$$\begin{aligned} \text{Inward Pressure (Pressure)} &= -0.159 \log A + 1.08 \\ &= -0.159 \log 24.5 + 1.08 = 0.86 \end{aligned}$$

$$\begin{aligned} \text{Outward Pressure (Suction)} &= 0.159 \log A - 1.16 \\ &= 0.159 \log 24.5 - 1.16 = -0.94 \end{aligned}$$

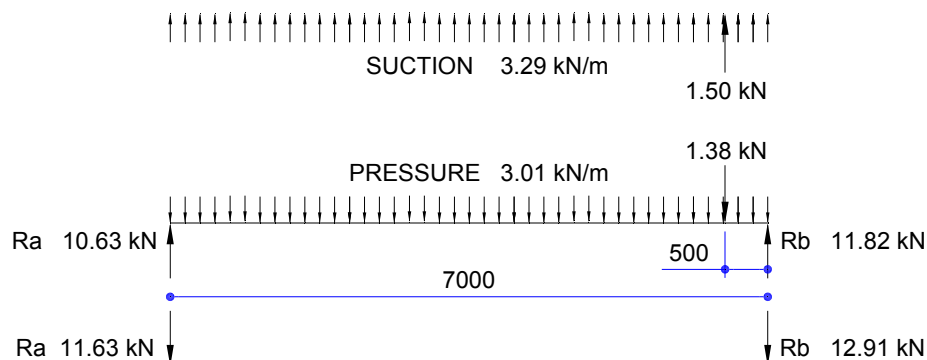
$$\text{Wind Load Pressure} = 1.00 \times 0.86 \times 3.5 = 3.01 \text{ kN/m}$$

$$\text{Wind Load Suction} = 1.00 \times -0.94 \times 3.5 = -3.29 \text{ kN/m}$$

Jamb is also supporting the header hence we need to consider the reactions from the header

$$\begin{aligned} &= 1.38 \quad \text{Pressure} \\ &= 1.50 \quad \text{Suction} \end{aligned}$$

Loading diagram is as below,



$$\begin{aligned}\text{Location of maximum bending moment pressure} &= 10.63 / 3.01 = 3.53 \text{ m} \\ \text{Location of maximum bending moment suction} &= 11.63 / 3.29 = 3.53 \text{ m}\end{aligned}$$

$$\text{Maximum Moment Pressure} = 10.63 \times 3.53 = 37.52 \text{ kNm}$$

$$\text{Maximum Moment Suction} = 11.63 \times 3.53 = 41.05 \text{ kNm}$$

The moment and shear capacities for 200C25 and, 2x200C25 are,

$$\begin{aligned}\text{Maxo} &= 11.93 \text{ kNm} &= 23.86 \text{ kNm} \text{ (Pressure)} \\ \text{Maxo2} &= 11.93 \times 0.65 = 7.75 \text{ kNm} &= 15.50 \text{ kNm} \text{ (Suction)} \\ \text{Vay} &= 46.59 \text{ kN} &= 93.18 \text{ kN}\end{aligned}$$

We see the capacities are much below required, even for double 'C' section, hence hot-rolled or built-up section may be used as jambs for above case.

b) Considering, the opening for a Sliding Door

Designing the door leaf members of size 3 x 6

1) DSD Inner Stile, Span = 1.50 m, Width = 1.50 m

Effective wind load area is the greater of,

$$\text{Tributary area} = 1.50 \times 1.50 = 2.25 \text{ m}^2$$

$$\text{Span} \times \text{Span} / 3 = 1.5 \times 1.5 / 3 = 0.75 \text{ m}^2$$

$$\text{Hence effective wind load area} = 2.25 \text{ m}^2$$

Considering an enclosed building and roof slope less than 10°, referring table 1.4.6 (a) of MBMA 02, GCp for interior zone is given by,

$$\begin{aligned}\text{Inward Pressure (Pressure)} &= -0.159 \text{ Log } A + 1.08 \\ &= -0.159 \text{ Log } 2.25 + 1.08 = 1.02\end{aligned}$$

$$\begin{aligned}\text{Outward Pressure (Suction)} &= 0.159 \text{ Log } A - 1.16 \\ &= 0.159 \text{ Log } 2.25 - 1.16 = -1.10\end{aligned}$$

$$\text{Wind Load Pressure} = 1.00 \times 1.02 \times 1.5 = 1.53 \text{ kN/m}$$

$$\text{Wind Load Suction} = 1.00 \times -1.10 \times 1.5 = -1.65 \text{ kN/m}$$

$$\text{Moment Pressure} = 1.53 \times 1.5^2 / 8 = 0.43 \text{ kNm}$$

$$\text{Shear Pressure} = 1.53 \times 1.5 / 2 = 1.15 \text{ kN}$$

$$\text{Moment Suction} = 1.65 \times 1.5^2 / 8 = 0.46 \text{ kNm}$$

$$\text{Shear Suction} = 1.65 \times 1.5 / 2 = 1.24 \text{ kN}$$

The compression flange is considered laterally braced by the sheeting under wind pressure loading, hence full moment capacity can be used. Where are for wind suction loading, the full moment capacity is reduced by applying the reduction factor R = 0.70

The moment and shear capacities for 120C20 are,

$$\begin{aligned} \text{Maxo} &= 4.54 \text{ kNm} & (\text{Pressure}) \\ \text{Maxo2} &= 4.54 \times 0.70 = 3.18 \text{ kNm} & (\text{Suction}) \\ \text{Vay} &= 28.78 \text{ kN} \end{aligned}$$

Therefore the stress ratios are,

Pressure	$M / \text{Maxo2}$	$= 0.43 / 4.54 = 0.09$	OK
	V / Vay	$= 1.15 / 28.78 = 0.04$	OK
Suction	M / Maxo	$= 0.46 / 3.18 = 0.15$	OK
	V / Vay	$= 1.24 / 26.89 = 0.05$	OK

Check Deflection,

$$I_{xx} = 1.19 \times 10^6 \text{ mm}^4$$

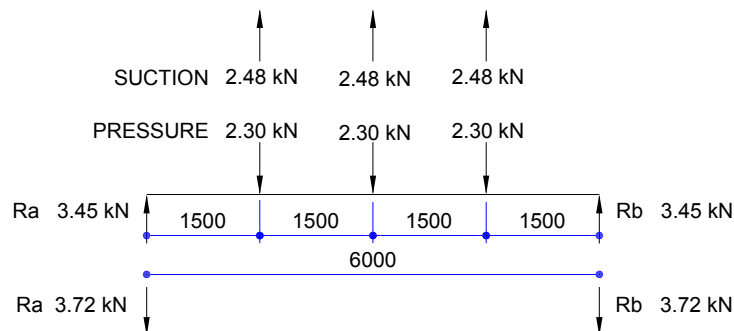
Calculated deflection, suction

$$\begin{aligned} &= 5 \times 1.65 \times 1000 \times 1500^4 / (384 \times 205000 \times 1.19 \times 10^6 \times 1000) \\ &= 0.45 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Allowable deflection} &= \text{span} / 90 = 1500 / 90 \\ &= 16.67 \text{ mm} \quad \text{OK} \end{aligned}$$

2) DSD Central Stile, Span = 6.0 m, Width = 1.50 m

The loads on the center stile is twice the reaction from inner stile, as shown below,



Moment Pressure	$= 3.45 \times 3.00 - 2.30 \times 1.5$	$= 6.90 \text{ kNm}$
Shear Pressure		$= 3.45 \text{ kN}$
Moment Suction	$= 3.72 \times 3.00 - 2.48 \times 1.5$	$= 7.44 \text{ kNm}$
Shear Suction		$= 3.72 \text{ kN}$

The compression flange is considered laterally braced by the inner stile framing under both loading wind pressure and suction loading, hence full moment capacity can be used.

The moment and shear capacities for 2x120C20 are,

$$\begin{aligned} \text{Maxo} &= 4.54 \times 2 = 9.08 \text{ kNm} \quad (\text{Pressure \& Suction}) \\ \text{Vay} &= 28.78 \times 2 = 57.56 \text{ kN} \end{aligned}$$

Therefore the stress ratios are,

Pressure	M / Maxo2	$= 6.90 / 9.08 = 0.76$	OK
	V / Vay	$= 3.45 / 57.56 = 0.06$	OK
Suction	M / Maxo	$= 7.44 / 9.08 = 0.82$	OK
	V / Vay	$= 3.72 / 57.56 = 0.06$	OK

Check Deflection,

$$I_{xx} = 2 \times 1.19 \times 10^6 \text{ mm}^4$$

Calculated deflection, suction

$$\begin{aligned} &= 0.0495 \times 2.48 \times 1000 \times 6000^3 / (205000 \times 2 \times 1.19 \times 10^6) \\ &= 54.34 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Allowable deflection} &= \text{span} / 90 = 6000 / 90 \\ &= 66.67 \text{ mm} \quad \text{OK} \end{aligned}$$

Edge Stile will be similarly designed for half the load of center stile.

3) DSD Bottom Stile, Span = 3.00 m, Width = 0.75 m

Effective wind load area is the greater of,

$$\text{Tributary area} = 0.75 \times 3.00 = 2.25 \text{ m}^2$$

$$\text{Span} \times \text{Span} / 3 = 3.0 \times 3.0 / 3 = 3.00 \text{ m}^2$$

$$\text{Hence effective wind load area} = 3.00 \text{ m}^2$$

Considering an enclosed building and roof slope less than 10° , referring table 1.4.6 (a) of MBMA 02, GC_p for interior zone is given by,

$$\begin{aligned} \text{Inward Pressure (Pressure)} &= -0.159 \log A + 1.08 \\ &= -0.159 \log 3.0 + 1.08 = 1.00 \end{aligned}$$

$$\begin{aligned} \text{Outward Pressure (Suction)} &= 0.159 \log A - 1.16 \\ &= 0.159 \log 3.0 - 1.16 = -1.08 \end{aligned}$$

$$\text{Wind Load Pressure} = 1.00 \times 1.00 \times 0.75 = 0.75 \text{ kN/m}$$

$$\text{Wind Load Suction} = 1.00 \times -1.08 \times 0.75 = -0.81 \text{ kN/m}$$

$$\begin{aligned} \text{Moment Pressure} &= 0.75 \times 3.0^2 / 8 + 3.45 \times 3.0 / 4 \\ &= 3.43 \text{ kNm} \end{aligned}$$

$$\text{Shear Pressure} = 0.75 \times 3.0 / 2 + 3.45 / 2 = 2.85 \text{ kN}$$

$$\begin{aligned} \text{Moment Suction} &= 0.81 \times 3.0^2 / 8 + 3.72 \times 3.0 / 4 \\ &= 3.71 \text{ kNm} \end{aligned}$$

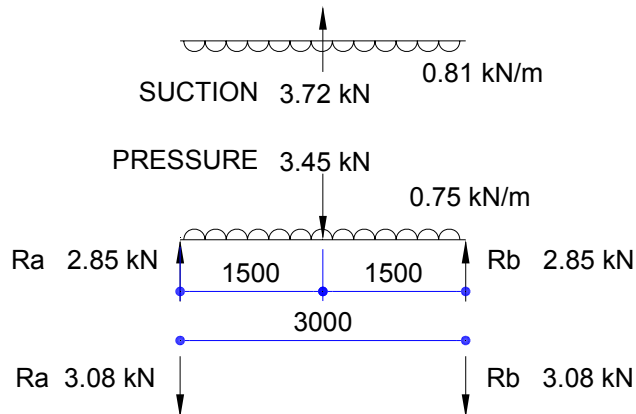
$$\text{Shear Suction} = 0.81 \times 3.0 / 2 + 3.75 / 2 = 3.09 \text{ kN}$$

The compression flange is considered laterally braced by the inner stile framing under both loading wind pressure and suction loading, hence full moment capacity can be used.

The moment and shear capacities for 120C20 are,

$$M_{axo} = 4.54 \text{ kNm (Pressure \& Suction)}$$

$$V_{ay} = 28.78 \text{ kN}$$



Therefore the stress ratios are,

Pressure	M / Maxo2	= 2.85 / 4.54	= 0.63	OK
	V / Vay	= 3.43 / 28.78	= 0.12	OK
Suction	M / Maxo	= 3.09 / 4.54	= 0.68	OK
	V / Vay	= 3.71 / 28.78	= 0.13	OK

Check Deflection,

$$I_{xx} = 1.19 \times 10^6 \text{ mm}^4$$

Calculated deflection, suction

$$\begin{aligned}
 &= 5 \times 0.81 \times 1000 \times 3000^4 / (384 \times 205000 \times 1.19 \times 10^6 \times 1000) \\
 &+ 3.72 \times 1000 \times 3000^3 / (48 \times 205000 \times 1.19 \times 10^6) \\
 &= 3.50 + 8.58 = 12.08 \text{ mm}
 \end{aligned}$$

Allowable deflection	= span / 90	= 3000 / 90	
	= 33.33 mm		OK

Similar check can be made for 250C20 and 250C25 when the girts are 250 mm deep. And 360M20 for 360 deep girts, note that the jambs and header for 360 deep sections are without the large oval holes.